

Spousal Homicide Rates in Ontario and Quebec, 1997–2023: Regional Patterns and the Relationship with Divorce

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Abstract

This paper investigates spousal homicide rates in Ontario and Quebec using crime and demographic data from Statistics Canada over the period 1997–2023. The analysis is twofold. First, we conduct a comparative analysis of spousal homicide rates between the two provinces, employing numerical summaries, box plots, histograms, time series, and normality tests to determine which province exhibits a higher incidence of spousal homicides. Second, we investigate the correlation between spousal homicide rates and divorce rates in Ontario by performing a linear regression, analyzing residuals, and evaluating the strength of the relationship through the correlation coefficient and coefficient of determination. Our comparative analysis reveals that Ontario exhibits a higher and more consistent spousal homicide rate, while Quebec displays greater variability. Our correlation analysis reveals a moderate positive correlation between divorce rates and spousal homicide rates in Ontario, though divorce rates alone account for only a small portion of the variance in spousal homicides. We discuss the limitations of these findings, suggest plausible explanations, and propose directions for future research.

1 Introduction

Spousal homicide remains a pressing human rights concern in Canada. Examining spousal homicide rates in two of Canada’s most populous provinces, Ontario and Quebec, can shed light on regional patterns of domestic violence and aid efforts to protect vulnerable individuals. A deeper understanding of these rates not only provides insight into regional variations but also highlights patterns in domestic abuse across the country.

This paper addresses two research questions. First, we ask: *which province, Quebec or Ontario, exhibits a higher incidence of spousal homicides?* By comparing the distributions, central tendencies, and temporal trends of spousal homicide rates in both provinces, we aim to identify meaningful differences between the two regions. Second, we ask: *is there a significant correlation between spousal homicide rates and divorce rates in Ontario?* Given that the spousal homicide data includes divorced victims, it is reasonable to anticipate some degree of correlation between spousal homicide and divorce rates.

We hypothesize a moderate positive correlation between divorce rates and spousal homicide rates, arising from heightened tensions during the process of divorce that may contribute to incidents of domestic violence escalating to manslaughter or murder. Furthermore, since only one party need agree to a divorce, the initiation of divorce proceedings may incite premeditated acts of violence in cases where one party reacts negatively. We note that other variables such as demographics, economic conditions, and access to social services are not explicitly controlled for in this analysis, and we discuss their potential influence in our conclusions.

2 Data and Methodology

2.1 Spousal Homicide Data

The spousal homicide data was obtained from Statistics Canada, covering the years 1997 to 2023 for both Ontario and Quebec. Throughout this paper, “spousal homicide” encompasses all Criminal Code offenses of murder, manslaughter, and infanticide, and includes incidents involving legally married, common-law, separated, divorced, and current and former same-gender spouses of victims aged 15 years or older.

For each province, the spousal homicide rate for a given year i is defined as

$$x_i = \frac{H_i}{P_i} \cdot 10^6,$$

where H_i represents the number of spousal homicides and P_i represents the population in year i . We analyze the rate per 1,000,000 people to address scaling issues, as the raw numbers of spousal homicides are relatively small. This approach ensures that even minor variations in homicide rates are captured effectively and allows for meaningful comparison between regions.

The homicide data was sourced directly from Ontario and Quebec police databases via the Homicide Survey, and therefore no sampling was involved, eliminating all forms of sampling bias. However, there is potential for measurement bias, as some murders may go unreported and instead be classified as missing persons cases. Population data, which includes Canadian citizens, permanent residents, and (since 1991) non-permanent residents, was extracted from administrative records. This may introduce potential under-coverage bias due to undocumented individuals and measurement bias due to difficulty accounting for inter-provincial migration.

2.2 Divorce Rate Data

The divorce rate data was obtained from the Canadian Vital Statistics – Divorce Database, which collects demographic information on divorce proceedings from the Central Registry of Divorce Proceedings. This data includes details such as court of registration, marriage and divorce dates, applicant information, and sex. The database includes all divorce proceedings in Canada rather than a sample, thus eliminating any potential sampling bias. However, there is potential measurement bias arising from couples who were divorced in another province and subsequently moved to Ontario, which may skew the data.

We analyze the divorce rate as the number of divorces per 1,000 married people. We choose to analyze divorce rates within the married population rather than the general population, as this subset is directly relevant to spousal homicides. This approach allows for a more direct comparison between divorce and spousal homicide data and reduces the influence of unrelated demographic changes in the general population.

2.3 Classification of Variables

Both the spousal homicide rate and divorce rate are quantitative and discrete, as they take on numerical values belonging to $\mathbb{Q}$ (the set of rationals), which is countable. Although the rates may appear continuous, they are expressed as ratios of natural numbers and are therefore discrete.

In the comparative analysis, the spousal homicide rate is the primary variable of interest. In the correlation analysis, we treat the divorce rate as the explanatory variable and the spousal homicide rate as the response variable, as we wish to investigate the effect of divorce rates on spousal homicides.

3 Comparative Analysis: Ontario and Quebec

We now compare spousal homicide rates in Ontario and Quebec over the period 1997–2023, using numerical summaries, box plots, histograms, time series, and normality tests. For brevity, we refer to a data point by its numerical value (e.g., 3.0) rather than writing “3.0 spousal homicides per 1,000,000 people” in full.

3.1 Numerical Summaries

	Min	Q1	Median	Mean	Q3	SD	Max	Range
Quebec	0.356	1.199	1.511	1.632	2.089	0.655	3.219	2.863
Ontario	0.952	1.288	1.668	1.626	1.795	0.510	3.344	2.392

Table 1: Numerical summaries of spousal homicide rates (per 1,000,000) in Quebec and Ontario, 1997–2023.

3.1.1 Central Tendency

The median spousal homicide rate in Quebec is 1.511, and the mean is slightly higher at 1.632. These two values are relatively close, indicating approximate symmetry; however, because the mean exceeds the median, higher values are pulling the center upward, suggesting a slight right skew. Ontario’s median spousal homicide rate is 1.668, and the mean is 1.626. The small difference between Ontario’s mean and median points to a more symmetrical distribution, and notably, Ontario’s center is slightly higher than Quebec’s.

3.1.2 Spread

Quebec’s standard deviation is 0.655 with a range of 2.863, while Ontario’s standard deviation is 0.510 with a range of 2.392. Since both Ontario’s standard deviation and range are smaller than Quebec’s, Ontario’s data is more concentrated around the center, and its spousal homicide rate varies less from year to year.

3.2 Box Plots

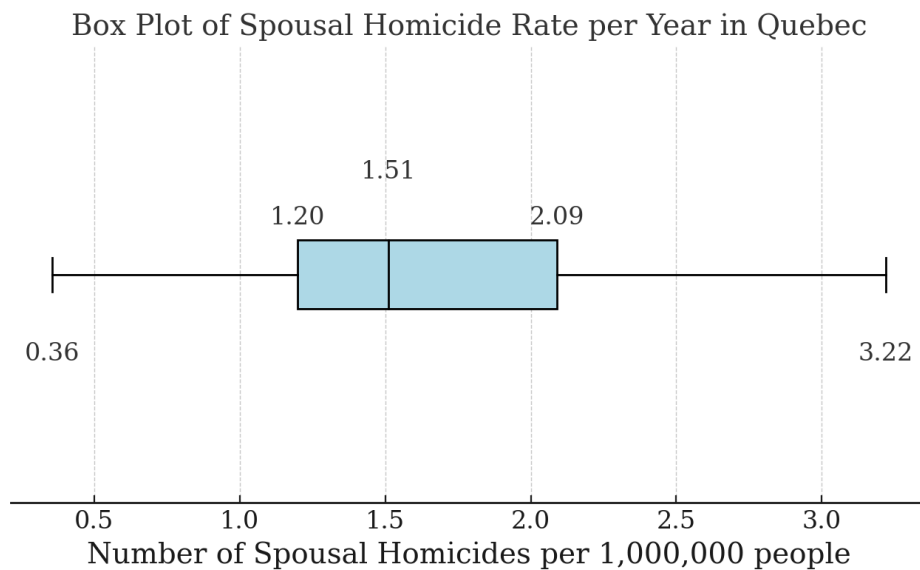


Figure 1: Box plot of spousal homicide rates in Quebec, 1997–2023.

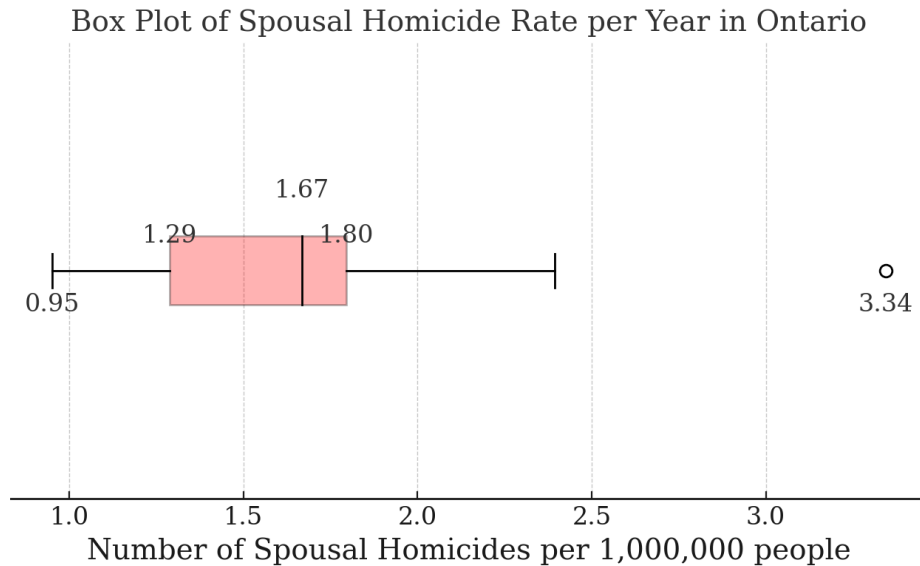


Figure 2: Box plot of spousal homicide rates in Ontario, 1997–2023.

3.2.1 Quartile Spread

The interquartile range (IQR) of Quebec’s data is 0.89, while the IQR for Ontario is 0.51. This confirms that Quebec’s spousal homicide rate exhibits more variability, whereas Ontario’s data is more concentrated around the center, consistent with our earlier observations regarding standard deviation and range.

3.2.2 Extrema

Quebec’s data has a minimum of 0.36 and a maximum of 3.22. Ontario’s minimum is 0.95 and its maximum is 3.34. While Quebec’s lower and upper bounds are each smaller than Ontario’s, this does not necessarily indicate that Quebec has lower spousal homicide rates overall. Ontario’s maximum is an outlier; if it is disregarded, the right whisker of its box plot extends only to approximately 2.5, indicating less variability in the data and further supporting the observation of a more consistent distribution in Ontario.

3.3 Histograms

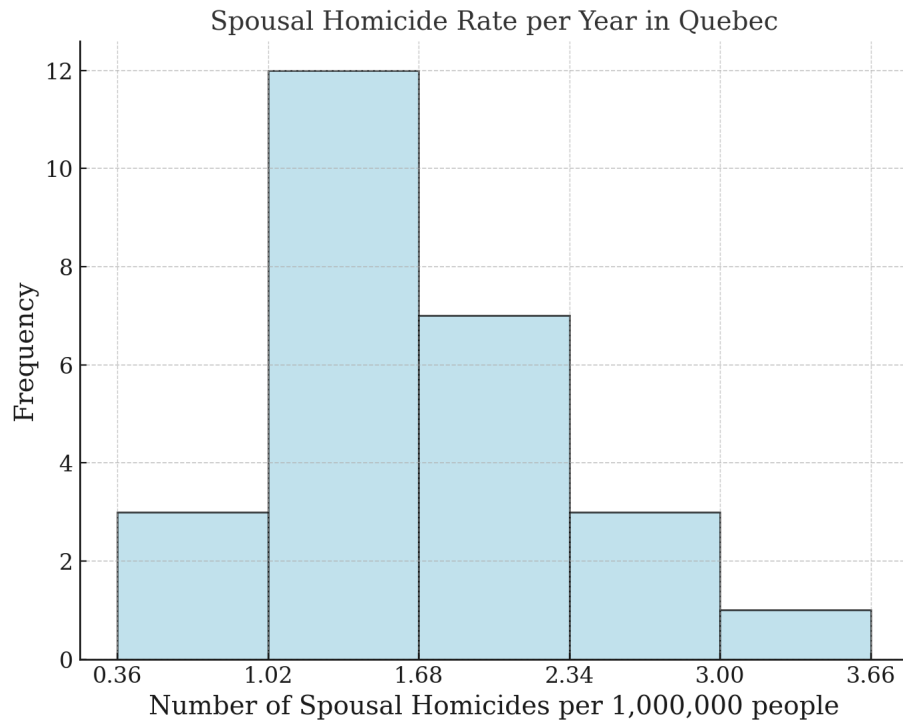


Figure 3: Histogram of spousal homicide rates in Quebec, 1997–2023.

The Quebec histogram has a unimodal distribution with a slight right skew and no unusual features. While a skew is present, the distribution appears approximately normal, supported by the majority of data points being concentrated around the central tendency, with both the mean and median residing within the peak interval $[1.02, 2.36]$. The frequencies on either side of this interval differ by only one, further suggesting approximate symmetry.

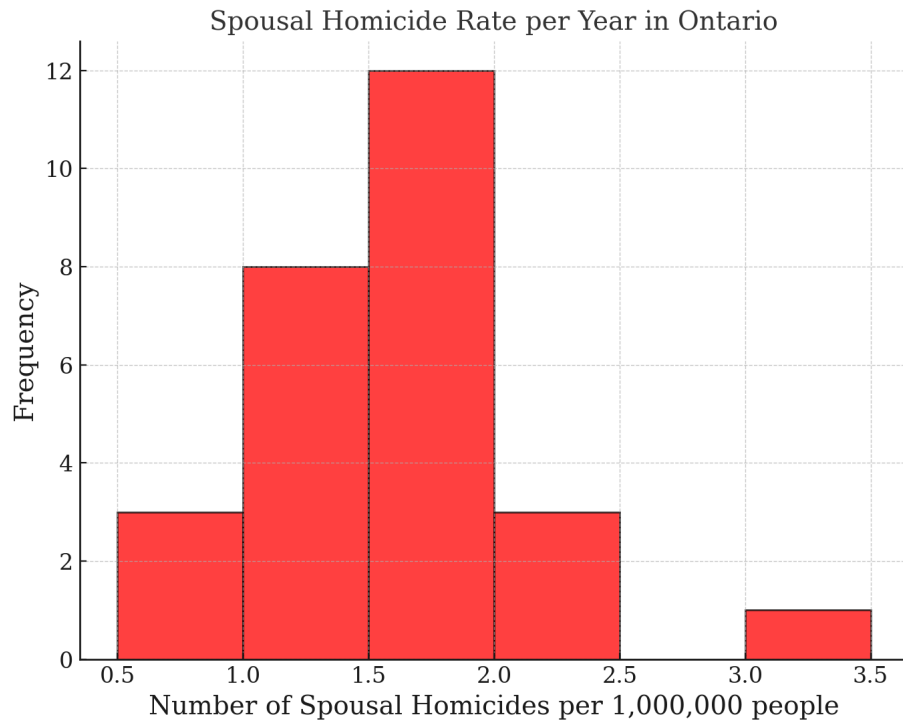


Figure 4: Histogram of spousal homicide rates in Ontario, 1997–2023.

The Ontario histogram has a unimodal distribution with a right skew, and the data tends to cluster around the central tendencies but more substantially on the left, indicating that below-average spousal homicide rates are more frequent. There is a notable gap in the data over the interval $[2.5, 3.0]$ due to the presence of a high value within $[3.0, 3.5]$. This gap signifies that the rates either remained below 2.5 or exceeded 3.0 across the observed years, with no intermediate values. If we exclude this outlier, the histogram more closely resembles a normal distribution.

3.4 Time Series

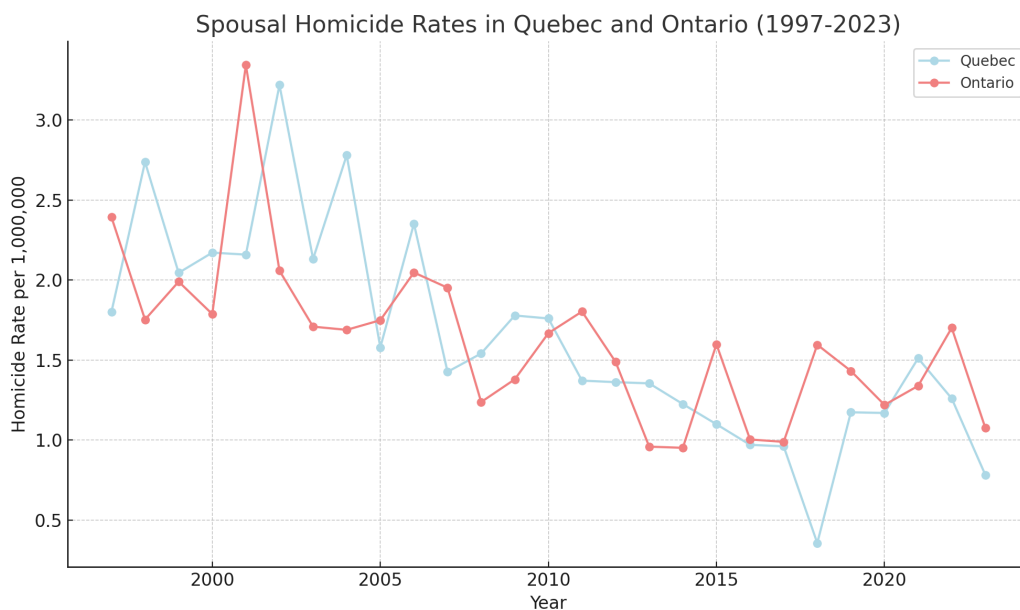


Figure 5: Time series of spousal homicide rates in Quebec and Ontario, 1997–2023.

3.4.1 Quebec

The spousal homicide rate in Quebec begins slightly above the median and mean in 1997 and grows in the following years, with three prominent peaks over the interval [1998, 2004] where rates surpassed 2.5, reaching a maximum above 3.0 in 2002. A significant drop occurs in 2005, followed by another peak in 2006. After 2006, the rates begin to stabilize and fluctuate less, especially over the years [2011, 2017], where there is a steady decrease with all rates falling below 1.5. The rates fluctuate again after reaching a minimum in 2018, increasing to 1.5 before falling below 1.0 in 2023. In summary, while there is considerable unpredictability due to fluctuations in the earlier years—consistent with Quebec’s wider standard deviation, higher variability, and larger range—there is a clear overall decrease in spousal homicide rates over the 26 years, with a more stable trend in the latter period.

3.4.2 Ontario

The time series for Ontario reveals a generally decreasing trend with less variability than Quebec. Starting around 2.4 in 1997, the rate decreases slightly in 1998 and experiences a minor increase in 1999. A significant spike occurs in 2001, where the rate reaches Ontario’s maximum—identified earlier as an outlier. After 2001, the rates tend to oscillate around the central tendencies while gradually decreasing, which aligns with Ontario’s lower standard deviation and variability. This downward trend continues into the later years, with spousal homicide rates often remaining below 1.5. Notably, in 2013 and 2014, the rates dip below 1.0, marking the lowest incidences during the observed period. In summary, Ontario’s spousal homicide rates decrease slowly and oscillate around the central tendencies with relatively low variability, consistent with our previous findings.

3.5 Assessing Normality

Since our data spans only 27 years, it is difficult to assess whether the distributions follow a normal distribution from histograms alone. We therefore employ the Shapiro–Wilk test and Q–Q plots to evaluate normality more rigorously.

3.5.1 Shapiro–Wilk Test

The Shapiro–Wilk test produces a p -value indicating whether data are consistent with a normal distribution. If $p > 0.05$, we do not reject the null hypothesis of normality; if $p \leq 0.05$, we reject it.

The p -value for Quebec is $0.52 > 0.05$, suggesting that Quebec’s spousal homicide rates are consistent with a normal distribution. For Ontario, the p -value is $0.0095 < 0.05$, leading us to reject normality. However, if we exclude the outlier at 3.344, the p -value increases to $0.38 > 0.05$, indicating that the non-normality is driven primarily by this single extreme observation.

3.5.2 Q-Q Plots

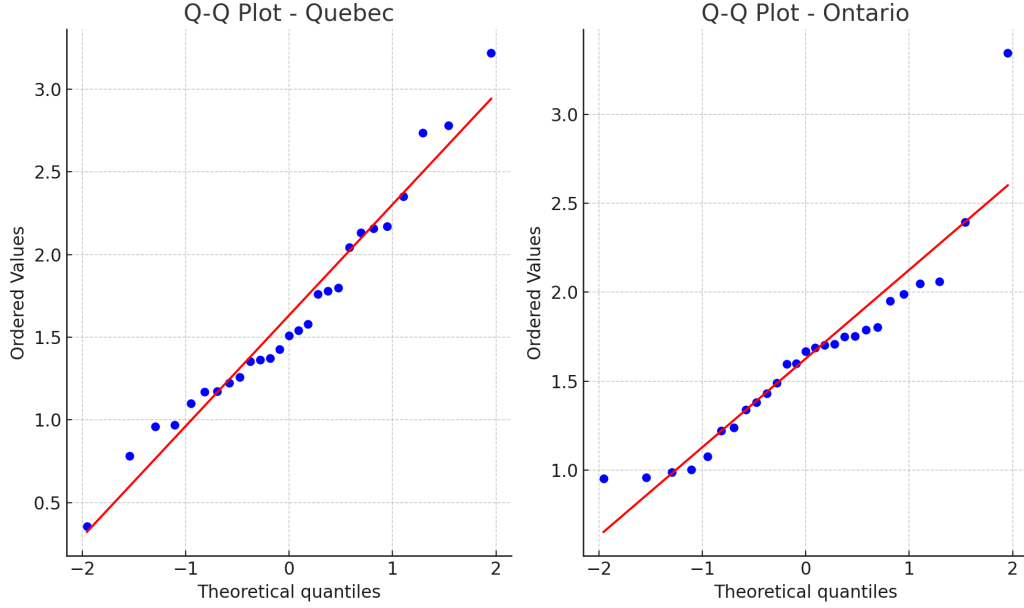


Figure 6: Q-Q plots for Quebec and Ontario spousal homicide rates, 1997–2023.

Quebec’s Q-Q plot shows that all data points lie relatively close to the reference line without substantial deviation, supporting approximate normality with a slight skew. Ontario’s Q-Q plot tends to deviate from the line, creating an S-shape that suggests non-normality. The outlier deviates significantly from the reference line; excluding it, the Q-Q plot exhibits an approximately normal pattern.

3.5.3 Normality Conclusions

Based on the Shapiro–Wilk test, Q-Q plots, and histogram shapes, Quebec’s spousal homicide rate distribution is approximately normal. Ontario’s distribution is rendered non-normal by the presence of an extreme outlier; excluding this outlier yields an approximately normal distribution.

3.5.4 Theoretical Probability Calculation

Assuming Ontario’s distribution (with the outlier excluded) is approximately normal, we compute the probability that a randomly selected spousal homicide rate falls within the interquartile range $[1.29, 1.80]$. Under the normal assumption, the probability density function is given by

$$\xi(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right),$$

and the probability of a data point falling in the interval $[a, b]$ is

$$P(a \leq x \leq b) = \int_a^b \xi(x) dx.$$

Substituting $\mu = 1.626$, $\sigma = 0.51$, $a = 1.29$, and $b = 1.80$, we obtain

$$P(1.29 \leq x_i \leq 1.80) = \int_{1.29}^{1.80} \frac{1}{0.51\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-1.626}{0.51}\right)^2\right) dx \approx 0.379.$$

Thus, under the normal assumption, there is approximately a 38% chance that a data point lies within the IQR. However, the empirical proportion of data points within the IQR is 48%, a notable discrepancy. This suggests that while the normal distribution provides a rough approximation, it does not perfectly characterize Ontario's spousal homicide rate distribution, and caution should be exercised when using it for probability calculations.

3.6 Comparative Conclusions

3.6.1 Measures of Central Tendency

For Quebec, the higher variability and the fact that the mean (1.632) exceeds the median (1.511) indicate that extreme high values are pulling the center upward, producing a slight right skew. Due to this skewness, the median is the more appropriate measure of central tendency for Quebec.

For Ontario, the mean (1.626) and median (1.668) are close, suggesting a relatively symmetrical distribution. Combined with the lower variability and tighter clustering observed in the box plots, histograms, and time series, both the mean and median are valid measures for Ontario. We adopt the mean for convenience in subsequent statistical calculations.

3.6.2 Which Province Exhibits Higher Spousal Homicide Rates?

Both Quebec and Ontario exhibit similar overall spousal homicide rates, with their centers lying relatively close to each other. Quebec displays greater variability, with higher and more frequent peaks in certain years and lower values in others. Ontario's rates, while more consistent with less variability, typically oscillate and cluster at higher levels than Quebec's, as evidenced by the box plots, histograms, and time series. Considering these factors along with Ontario's higher central tendency, we conclude that **Ontario exhibits a higher rate of spousal homicides** over the period 1997–2023.

4 Correlation Analysis: Divorce Rates and Spousal Homicide in Ontario

Having established the comparative landscape of spousal homicide rates, we now investigate whether a correlation exists between divorce rates and spousal homicide rates in Ontario. This analysis covers the period 1997–2020, as divorce rate data was available through 2020 at the time of analysis.

4.1 Scatter Plot and Correlation

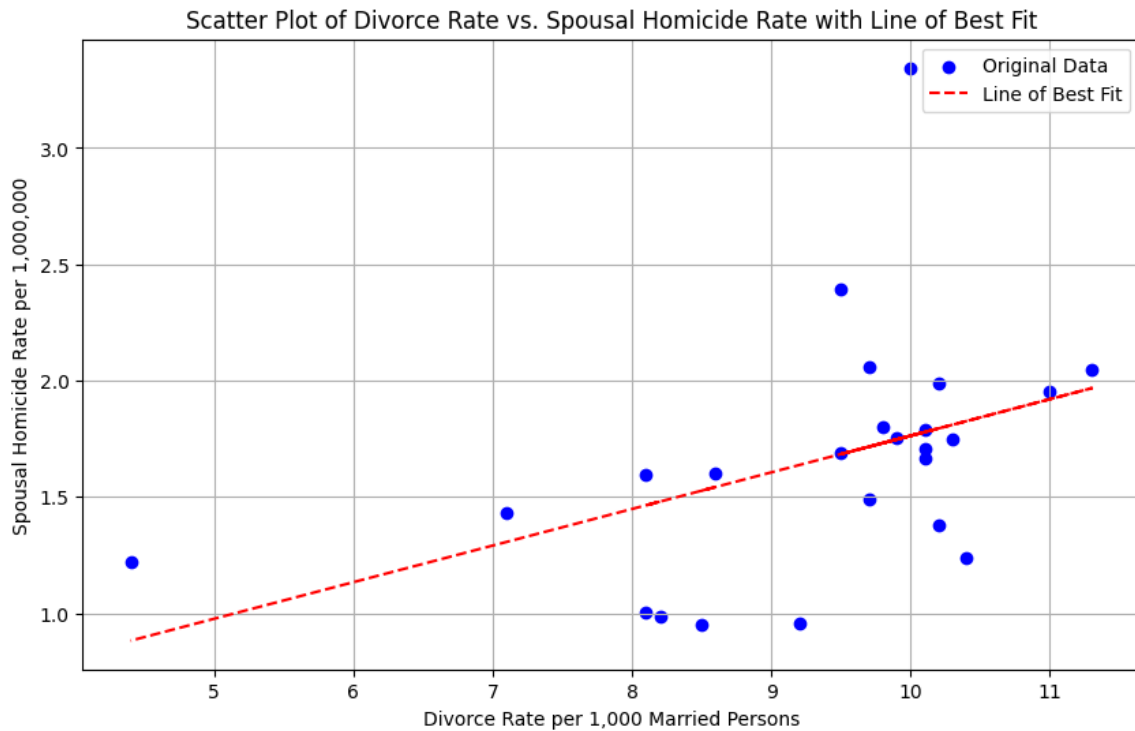


Figure 7: Scatter plot of divorce rate versus spousal homicide rate in Ontario, 1997–2020, with line of best fit.

Correlation Coefficient	Coefficient of Determination
$r = 0.433$	$r^2 = 0.188$

Table 2: Correlation statistics for the original dataset.

The correlation coefficient of approximately 0.433 indicates a moderate positive correlation between divorce rates and spousal homicide rates. This is reinforced by the scatter plot, which shows a loosely aligned positive trend: as divorce rates increase, spousal homicide rates tend to increase as well, though the relationship is not tightly clustered around a single line. This is expected, since spousal homicides do not solely involve divorced victims, and other variables such as demographics and economic conditions may also contribute to spousal homicide rates.

The coefficient of determination of approximately 0.188 indicates that 18.8% of the variance in spousal homicide rates can be explained by the divorce rate, and that the linear model is a weak predictor of spousal homicide rates.

4.2 Residual Analysis

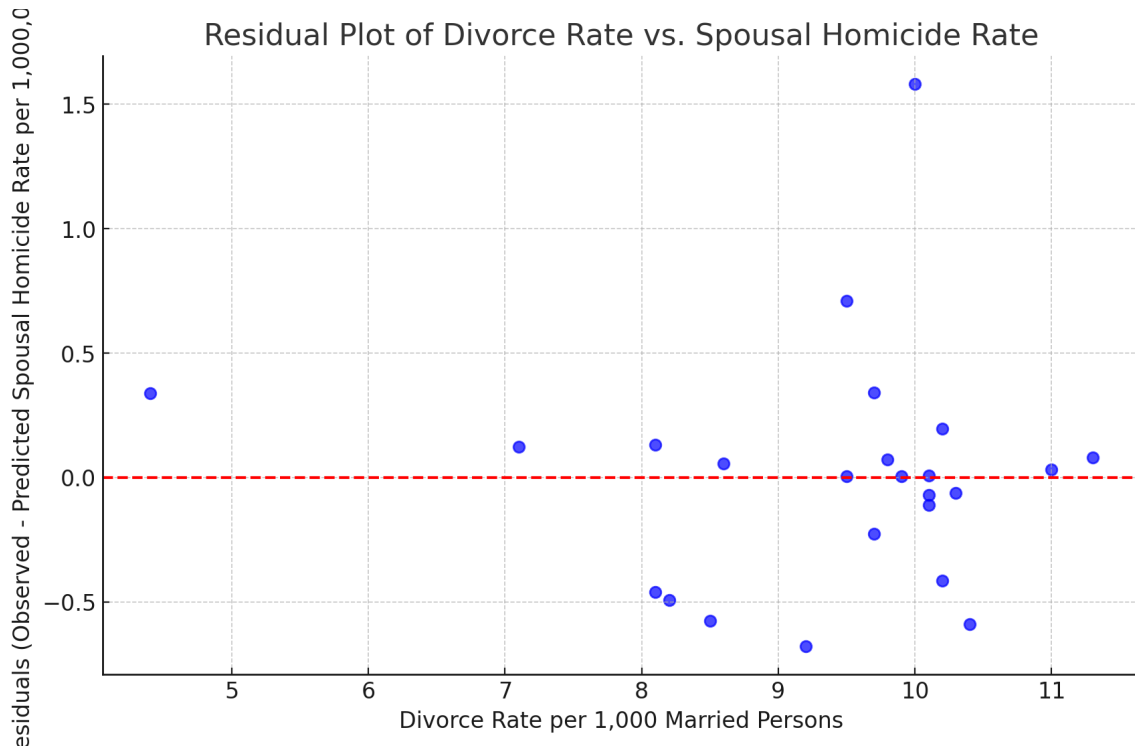


Figure 8: Residual plot for the linear model fitted to the original dataset.

To assess the appropriateness of the linear model, we examine the residual plot, which displays the difference between the observed spousal homicide rate and the value predicted by the linear model.

The distribution of residuals appears fairly random without obvious systematic patterns. However, at $x = 10$ there is a clear outlier lying beyond two standard deviations from zero, and some larger deviations appear in the interval $[8, 11]$. The outlier corresponds to a spousal homicide rate of 3.344 per 1,000,000, which occurred in 2001—the same year identified as an outlier in our comparative analysis. This spike may be related to the small recession caused by the collapse of the dot-com bubble, which could have had economic effects on Ontario and driven up financial tensions between spouses. We note this as a speculative explanation, particularly since the 2008 recession did not produce a comparable spike in spousal homicide rates. Apart from this outlier and some moderate deviations, the spread is reasonably random, suggesting that a linear model is an appropriate functional form for the data.

4.3 Outlier Removal

Removing the 2001 outlier, the updated correlation statistics are:

Correlation Coefficient	Coefficient of Determination
$r = 0.504$	$r^2 = 0.254$

Table 3: Correlation statistics with the 2001 outlier removed.

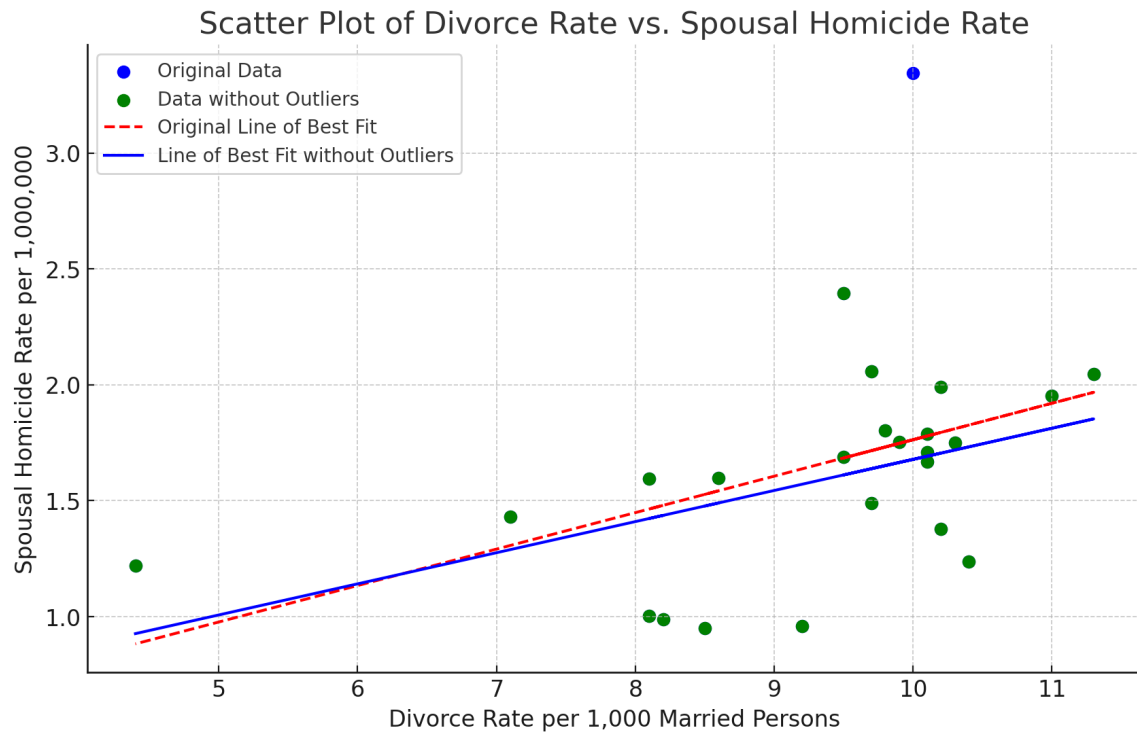


Figure 9: Scatter plot with line of best fit after removing the 2001 outlier.

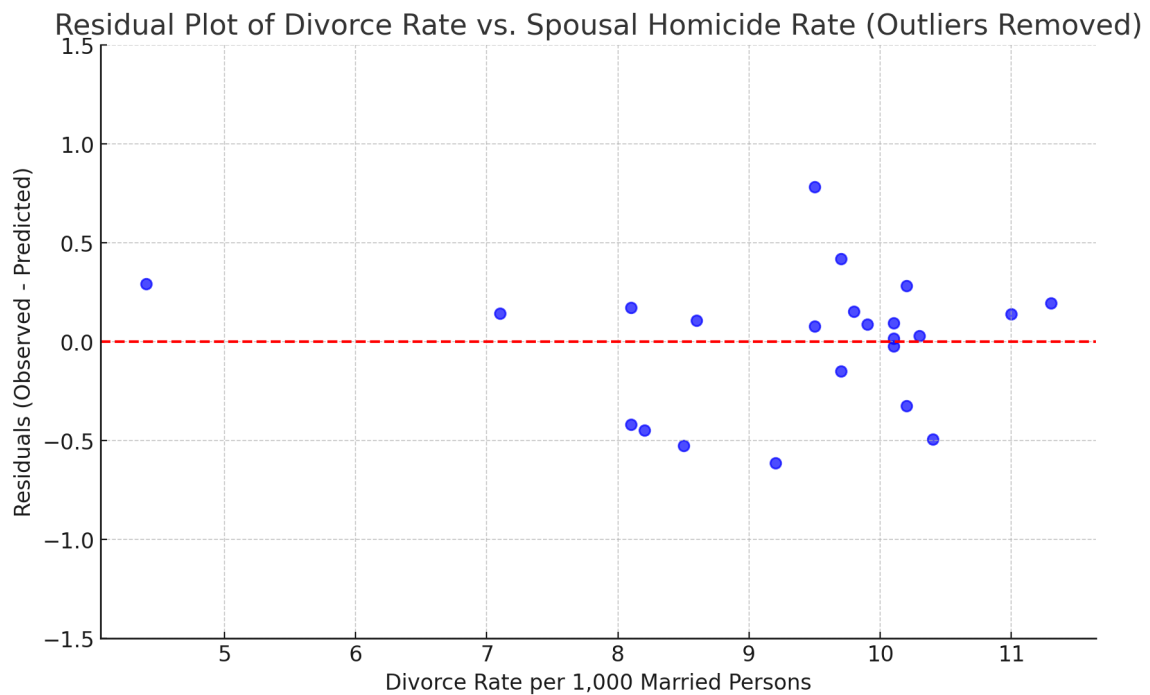


Figure 10: Residual plot after removing the 2001 outlier.

With the outlier removed, the correlation coefficient increases to approximately 0.504, representing a stronger moderate positive correlation compared to the original value of 0.433. The coefficient of determination increases to approximately 0.254, meaning that about 25.4% of the variance in spousal homicide rates can be explained by the divorce rate, compared to 18.8% with the outlier included. The residual plot after removing the outlier shows a more even distribution

around zero with less pronounced deviations, while maintaining the random pattern expected under a well-fitted linear model.

This outlier was a significant deviation, lying more than two standard deviations from the regression line, with the nearest data point at 2.393—a gap of nearly 1.0 spousal homicides per 1,000,000, which is substantial in the context of this dataset. The outlier likely represents a unique event driven by factors not captured by divorce rates alone, such as Ontario’s political or economic conditions in 2001. By removing it, the model aligns more closely with the assumptions of linear regression, yielding a more statistically valid estimate of the relationship between spousal homicide and divorce rates.

4.4 Linear Regression

We perform a linear regression on both the original and modified datasets. Our line of best fit takes the form $y(x) = ax + b$, where x denotes the divorce rate and y denotes the predicted spousal homicide rate. We seek to minimize the sum of squared residuals,

$$f(a, b) = \sum_{n=1}^N (y_n - ax_n - b)^2,$$

by solving the system

$$\frac{\partial f}{\partial a} = 0, \quad \frac{\partial f}{\partial b} = 0.$$

For the original dataset ($N = 20$), we obtain $a = 0.1573$ and $b = 0.1903$, giving the regression line

$$y(x) = 0.1573x + 0.1903.$$

For the modified dataset with the 2001 outlier removed ($N = 19$), we obtain $a = 0.3360$ and $b = 0.1342$, giving the regression line

$$y(x) = 0.3360x + 0.1342.$$

The steeper slope in the modified model indicates a stronger estimated relationship between divorce rates and spousal homicide rates when the anomalous 2001 observation is excluded.

4.5 Predictions

Using both regression models, we generate predicted spousal homicide rates for a range of divorce rate values:

5 Conclusions and Interpretations

5.1 Summary of Findings

Our analysis yields two principal findings. First, Ontario exhibits a higher and more consistent spousal homicide rate than Quebec over the period 1997–2023. While the central tendencies of the two provinces are similar, Ontario’s rates cluster higher with less variability, whereas Quebec shows greater dispersion with more pronounced peaks and troughs. Second, there is a moderate positive correlation between divorce rates and spousal homicide rates in Ontario, with $r = 0.433$ in the original dataset increasing to $r = 0.504$ when a significant outlier from 2001 is removed.

Divorce Rate	Predicted Homicide Rate	Predicted (Without Outlier)
4.0	0.819	0.873
4.5	0.898	0.940
5.0	0.977	1.007
5.5	1.055	1.074
6.0	1.134	1.141
6.5	1.213	1.208
7.0	1.292	1.275
7.5	1.370	1.342
8.0	1.449	1.409
8.5	1.528	1.476
9.0	1.606	1.543
9.5	1.685	1.610
10.0	1.764	1.677
10.5	1.843	1.744
11.0	1.921	1.811
11.5	2.000	1.878
12.0	2.079	1.945

Table 4: Predicted spousal homicide rates (per 1,000,000) based on divorce rates (per 1,000 married people).

The correlation between divorce rates and spousal homicide rates is best characterized as a common-cause relationship rather than a direct cause-and-effect one. Spousal homicide does not exclusively involve divorced victims, and it is therefore reasonable that divorce does not directly cause spousal homicides. Rather, individuals experiencing divorce may be subject to stressors and situational challenges that could heighten risks in volatile relationships. The 2001 outlier, where spousal homicide rates were significantly elevated, may correspond to external economic pressures such as the aftermath of the dot-com bubble collapse, which could have intensified financial and emotional strains independently of divorce rates.

5.2 Plausible Explanations

Factor	Explanation
Economic and Social Stressors	Economic hardship may increase the likelihood of both divorce and domestic violence, as financial strain often intensifies interpersonal tensions. Economic factors such as housing costs, food prices, and median household income can contribute to financial stress within households, raising tension and potentially escalating into domestic violence.
Mental Health and Conflict Resolution	Limited access to mental health support during divorce may hinder healthy coping mechanisms, potentially escalating conflict to violence. Social stigma around divorce could compound stress, increasing risks in already strained relationships.
Population Diversity	Ontario has a more diverse population with a higher proportion of immigrants. Cultural differences may contribute to varying rates of domestic violence across communities.
Urbanization	Ontario has a larger proportion of its population living in major metropolitan areas. Urban areas are generally associated with higher crime rates, including domestic violence.
Legal and Reporting Differences	Differences in law, law enforcement practices, or reporting standards between the provinces may affect the recorded spousal homicide rates.

Table 5: Plausible explanations for observed patterns.

5.3 Limitations

There are several limitations to our findings. The coefficient of determination $r^2 = 0.188$ (increasing to $r^2 = 0.254$ without the outlier) indicates that divorce rates account for only a small portion of the variance in spousal homicide rates and therefore have limited predictive power. External factors such as economic stability, access to mental health support, substance abuse, and demographics may significantly affect both divorce and spousal homicide rates, yet these factors are not controlled for in our analysis, which may obscure or confound the true relationship.

Measurement bias is present across several dimensions of the data. Spousal homicide rates may be lower than the true rates due to under-reporting. Population estimates may be inaccurate if undocumented individuals are not captured, causing spousal homicide rates to appear artificially higher or lower depending on whether these individuals contribute to homicide counts. Divorce rates may appear slightly lower due to under-coverage bias. Additionally, the comparative analysis between Ontario and Quebec could be affected by differences in law enforcement practices and reporting standards between the two provinces.

Finally, our findings are specific to Ontario and Quebec from 1997 to 2023 and may not generalize to other regions or time periods.

6 Next Steps in Research

Research Approach	Description
Multivariable Analysis	Incorporate additional variables such as unemployment rates, mental health data, substance abuse statistics, and demographic factors to clarify the underlying factors influencing both divorce and spousal homicide rates.
Qualitative Research	Conduct case studies or interviews with social workers, law enforcement, and mental health professionals to gain insight into situational factors contributing to spousal homicide cases, especially those involving recent separations.
Comparative Studies	Expand the research timeframe and compare data across additional provinces or countries to determine if the observed patterns are consistent across different contexts or specific to Ontario and Quebec during this period.
Urban vs. Rural Break-down	Compare spousal homicide rates in urban and rural areas within each province to reveal patterns related to population density and community support structures.
Investigate Contributing Factors to Divorce	Examine additional factors that may influence divorce rates, such as financial stress, family dynamics, and mental health, to better understand how these may relate to risks of spousal conflict or violence and to identify potential preventive measures.
Improved Social Services	Investigate the effect of providing greater access to financial and social services for households as a means of reducing stress and potentially lowering spousal homicide rates.

Table 6: Proposed next steps in research.

7 Raw Data

Year	Quebec	Ontario
1997	1.800	2.394
1998	2.738	1.753
1999	2.045	1.990
2000	2.171	1.788
2001	2.159	3.344
2002	3.219	2.058
2003	2.132	1.709
2004	2.780	1.689
2005	1.579	1.749
2006	2.352	2.047
2007	1.426	1.952
2008	1.541	1.238
2009	1.779	1.380
2010	1.760	1.668
2011	1.371	1.803
2012	1.362	1.489
2013	1.354	0.959
2014	1.225	0.952
2015	1.099	1.599
2016	0.970	1.004
2017	0.961	0.989
2018	0.356	1.596
2019	1.174	1.432
2020	1.169	1.220
2021	1.511	1.339
2022	1.260	1.701
2023	0.782	1.076

Table 7: Spousal homicide rates (per 1,000,000) in Quebec and Ontario, 1997–2023.

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